

AI Resume Builder using Prompt Engineering

Project Synopsis

Abstract

This project is an AI-powered Resume Builder developed using Python and Streamlit. It uses Prompt Engineering with the OpenAI API to generate professional resume content from user inputs. The application helps users quickly create well-structured resumes with AI-generated summaries, skills, experience descriptions, and project details.

Objectives

Automate resume creation; Generate ATS-friendly content; Demonstrate Prompt Engineering; Provide an easy-to-use interface.

Technologies

Python, Streamlit, OpenAI API, Prompt Engineering, Environment Variables (.env), VS Code.

Modules

User Input Form, Prompt Generator, OpenAI Integration, Resume Preview, Download Option.

Working

The user enters personal, educational, and professional details. The application builds a structured prompt and sends it to the OpenAI API. The AI returns polished resume content, which is displayed and can be downloaded.

Advantages

Saves time, improves resume quality, produces professional language, customizable, beginner-friendly.

Applications

Job applications, internships, campus placements, freelance profiles, career transitions.

Future Enhancements

Multiple templates, PDF formatting, LinkedIn import, multilingual support, ATS score checker.

Conclusion

The project demonstrates how Prompt Engineering and Generative AI can automate resume writing while producing professional, structured, and personalized content.